

CYBER SECURITY

UNIT 5 – COMPLETE DETAILED NOTES

Proxy Servers & Anonymizers • Password Cracking • Keyloggers & Spyware • Virus & Worms • Trojan Horses • Backdoors

DoS & DDoS Attacks • Buffer Overflow • Attack on Wireless Networks • Phishing (Method & Techniques)

Book-style notes with Practical Examples, Definitions, Flowcharts, Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 5 – SYLLABUS (As per Examination)

S.No	Unit 5 Topics	Marks
1	Proxy Servers and Anonymizers	---
2	Password Cracking	---
3	Keyloggers and Spyware	---
4	Virus and Worms	---
5	Trojan Horses	---
6	Backdoors	---
7	DoS and DDoS Attacks	---
8	Buffer and Overflow (Buffer Overflow)	---
9	Attack on Wireless Networks	---
10	Phishing: Method of Phishing, Phishing Techniques	---

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

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PART 1 – PROXY SERVERS AND ANONYMIZERS

Cyber criminals apni **asli pehchan (IP)** chhupane ke liye **proxy server** aur **anonymizer** use karte hain — jaise chor mask pehen ke chalta hai.

1.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek hacker **Mumbai** me baitha tha. Usne ek bank website par attack kiya. Police ne **IP address** se pakadne ki koshish ki.

Par hacker ne pehle hi proxy server use kar liya tha:

- Uska request **proxy server** se hokar gaya — bank ko **proxy ka IP** dikha, hacker ka nahi.
- Proxy bhi kisi **doosre desh** me thi — trail toot gayi.
- Police ko pakadne me **bahut time laga**.

Isi wajah se proxy/anonymizers cyber crime me itne common hain.

1.2 Definition

DEFINITION

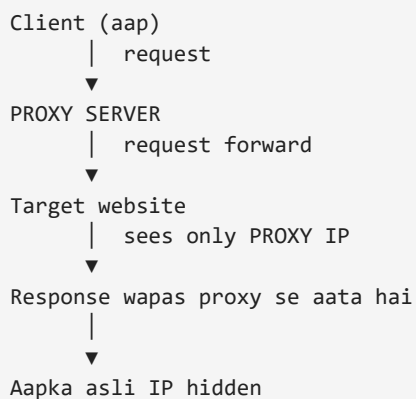
Proxy Server — beech ka server jo client (aap) aur internet ke darmiyaan request forward karta hai — aapka **asli IP server se chhupa** jaata hai.

Anonymizer — tool/service jo aapki **identity aur IP ko fully chhupati** hai — surfing par koi pata na chale ki aap kaun hain.

Simple: Proxy = **pehredar ka dikhna** (wo request aage badhata hai), Anonymizer = **inka baap** — identity bilkul chhupa.

1.3 Flowchart

FLOWCHART



1.4 Proxy vs Anonymizer vs VPN (Table)

Basis	Proxy Server	Anonymizer	VPN
IP chhupata hai	Haan (web traffic)	Haan (identity)	Haan (sab traffic)
Encryption	Aksar nahi	Hamesha nahi	Haan (encrypted)
Kaam	Request forward	Identity hide	Secure tunnel
Use (criminals)	IP chhupana	Identity chhupana	Fully anonymous

1.5 Kyu Use Kiya Jata Hai

- Cyber criminal **IP hide** karne ke liye — pakda na jaye.
- **Bypass restrictions** — blocked sites tak pahuchna.
- **Multi-layer proxy** (chain) — trail aur bhi mushkil.
- **Anonymizer** se email/chat bhi bina pehchan ke.
- Legal use bhi hai — privacy, corporate filtering.

1.6 Prevention (Bachao Ke Tarike)

- Investigators ko **proxy logs + ISP** se peechna padta hai — jaaldi karo.
- Multi-hop trails ke liye **timestamps + packet analysis**.
- Websites par **strong logging + anomaly detection**.
- Real-time **threat intel** se proxy-IP pools block karo.
- Companies me **reverse proxy** use karke attack block karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Proxy server = beech ka server — asli IP chhupata hai.
2. Anonymizer = identity/IP fully chhupane wala tool.
3. Criminal IP hide karne ke liye inka use karta hai — trail toot jati hai.
4. VPN = encrypted tunnel (sabse secure).

Viva Q1. Proxy server kya karta hai?

Ans: Client aur internet ke beech request forward karta hai — target ko asli IP nahi dikhta.

Viva Q2. Anonymizer ka kya kaam?

Ans: User ki identity/IP ko chhupana taaki pata na chale.

MEMORY TRICK

Proxy = Pehredar, Anonymizer = Inka baap. Trick: **Proxy = Paisa ka bichauliya** — jaise bichauliya aapke naam nahi batata, waise proxy aapka IP nahi batata.

PART 2 – PASSWORD CRACKING

Password Cracking — kisi ke account ka password nikalne ka tarika. Criminals iske liye weak passwords aur hash cracking tools use karte hain.

2.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek aadmi ka **Instagram account** hack ho gaya. Password tha — **123456**.

Hacker ne:

- Pehle **common passwords** try kiye (dictionary attack).
- Phir har letter-digit-combo **brute force** se try kiya.
- 123456 ek hi try me mil gaya.

Hacker ne account se paisa maangne wale messages bheje — aur account waapas karne ke liye paisa maanga.

Weak password hi sabse bada darwaza hai.

2.2 Definition

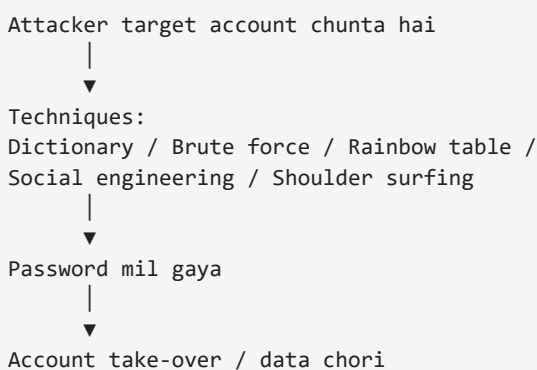
DEFINITION

Password Cracking — kisi user ke account ya system ka **password recover/guess** karne ka process — weak passwords, hash attacks, aur social engineering ke through.

Simple: Tala hai — par chabi itni kamzor ki koi bhi tod le.

2.3 Flowchart

FLOWCHART



2.4 Techniques (Table)

Technique	Kya hota hai
Dictionary attack	Common words/passwords ki list try karna (password, 123456)
Brute force	Har possible combo try karna — slow par full
Rainbow table	Pehle se bani hash table se password dhoondna
Credential stuffing	Doosri site ka leaked password yahan try karna
Social engineering	Logon se password poochna (call/message)
Shoulder surfing	Kandhe se dekh kar password yaad karna
Keylogging	Keylogger se password capture

2.5 Important Points

IMPORTANT POINTS

1. **Weak password** = seconds me toot jata hai.
2. **Hash attacks** — jab password hash me save hota hai — rainbow table se toota jata hai.
3. **Salt** (hash me random value) daalne se rainbow table bekaar ho jata hai.
4. **2FA** (OTP) hone par password ke baad bhi doosra guard rehta hai.
5. Criminals sabse pehle **common + leaked** passwords try karte hain.

2.6 Prevention (Bachao Ke Tarike)

- Password **12+ characters** — letters + numbers + symbols.
- **Alag-alag site** ke liye alag password (password manager).
- **2FA / OTP** har jagah ON karo.
- **Common passwords** (123456, password, qwerty) kabhi mat use karo.
- Leaked password check karo (haveibeenpwned jaisi sites).
- Cracking risk kam karne ke liye **account logout** + rate limiting.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Password cracking = password recover/guess karna.
2. Techniques = dictionary, brute force, rainbow table, social engineering, shoulder surfing.
3. Salt daalne se rainbow table kaam nahi karti.
4. Bachao = strong password + 2FA + alag-alag password.

Viva Q1. Brute force aur dictionary attack me farak?

Ans: Brute force har combo try karta hai; dictionary sirf common words/list try karta hai.

Viva Q2. Rainbow table kya hai?

Ans: Pehle se bani hash-value table jisse password turant dhoondha jata hai.

MEMORY TRICK

Password Cracking = Chabi ka todna. Trick: **Password = PassWORD** — jaise ghar ka darwaza strong lock, waise hi account strong password. 123456 = **khula darwaza**.

PART 3 – KEYLOGGERS AND SPYWARE

Keylogger har key (keystroke) record karta hai. **Spyware** aapki activities par nazar rakhta hai. Dono **chup-chaap** kaam karte hain aur passwords/data nikalte hain.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek cyber cafe me hacker ne **keylogger software** install kar diya.

- Jaise hi koi logon ne **email/bank** me login kiya — uski har **key press record** ho gayi.
- Hacker ko saare **usernames + passwords** mil gaye.
- Usne account se paisa nikal liya — jab tak victim ko pata chalta, nuksan ho chuka tha.

Cyber cafe ke computer par **spyware** bhi tha jo screen capture karta tha.

Keylogger (keys) + Spyware (activities) — dono chhup ke data churate hain.

3.2 Definition

DEFINITION

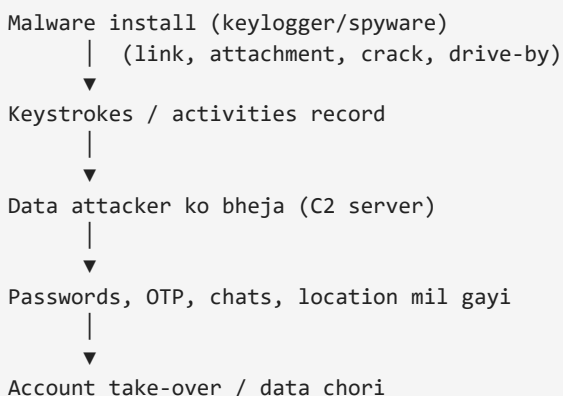
Keylogger — software/hardware jo user ki **har keystroke** record karta hai — password, OTP, messages sab capture ho jate hain.

Spyware — software jo **user ki activities par spy** karta hai — browsing, screenshots, camera, location, data — aur attacker ko bhejta hai.

Simple: Keylogger = **tap kiya hua phone** (typing sunai de); Spyware = **pichchhu** (sab dekh kar bata raha).

3.3 Flowchart

FLOWCHART



3.4 Types (Table)

Type	Kya karta hai
Software keylogger	App/rootkit se keys record karta hai
Hardware keylogger	USB/device jo keyboard se juda hota hai
Screen capture	Screenshots/screen record karta hai
Browser spyware	History, cookies, logins churata hai
Mobile spyware	SMS, calls, location, camera
Banking trojan	Bank pages par fake overlay banata hai

3.5 Keylogger vs Spyware (Table)

Basis	Keylogger	Spyware
Record karta hai	Keystrokes (typing)	Activities (browsing, screen, camera)
Target	Passwords, messages	Sab kuch (data, habits)
Detection	Mushkil (chhupa hota)	Antivirus se kabhi pakda jata
Goal	Login/password chori	Jasusi (espionage)

3.6 Prevention (Bachao Ke Tarike)

- **Antivirus + anti-spyware** install rakho (updated).
- Cyber cafe / public computer par **sensitive login na karo**.
- **Unknown attachments/links** na kholo.
- Apps **official store** se hi install karo.
- **2FA** — keylogger se password chori par bhi OTP guard.
- Task Manager/process list se **suspicious process** check karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Keylogger = har keystroke record karta hai (password chori).
2. Spyware = activities par spy (browsing, screen, camera).
3. Install = link/attachment/crack se; detection mushkil.
4. Bachao = antivirus + 2FA + public computer par login na karo.

Viva Q1. Keylogger ka target kya hota hai?

Ans: Passwords, OTP, messages — jo typing se aate hain.

Viva Q2. Spyware se kya kya pata chalta hai?

Ans: Browsing, screenshots, camera, location, data — sab.

MEMORY TRICK

Keylogger = Typing ka chhalla, Spyware = Screen ka bhediya. Trick: Keylogger me **KEY** hota hai — keyboard ki key. Jaise aap type karo, usne note kar liya.

PART 4 – VIRUS AND WORMS

Virus aur **worm** dono malware hain — par virus ko **host file chahiye**, worm **khud hi failta** hai network me. Ye computer ko slow, corrupt aur hackable banate hain.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Office me ek employee ko **email ke sath attachment** mila — 'salary_details.xls'. Usne khola — file me **virus** tha.

- Virus **attach hua** files se — document, pictures corrupt ho gaye.
- Har file khulte hi doosri file me virus copy ho gaya.

Kuch hafta pehle hi usi office me ek **worm** tha — jisne **network me khud fail kar** saare computers par emails bheji.

Result:

- Virus = files/folders kharab + backup me bhi chhupa.
- Worm = network saturated, internet slow, sab systems infected.

Virus = host se chipakta hai; Worm = khud bhagta hai network par.

4.2 Definition

DEFINITION

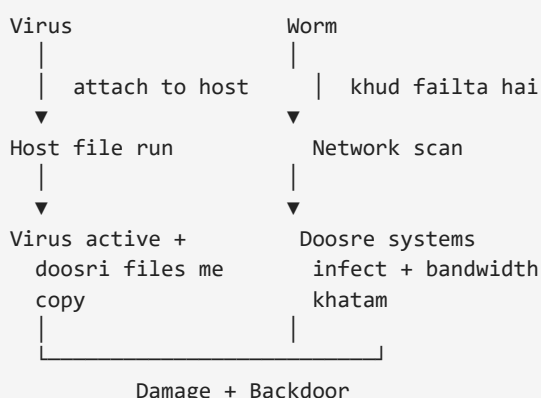
Virus — malware jo **host file/program me attach** hota hai aur jab file run hoti hai to failta hai — files corrupt, slow, damage.

Worm — **self-replicating** malware jo bina host ke **network me khud failta** hai — system slow, bandwidth khatam, backdoor bhi kholta hai.

Simple: Virus = **chipku** (file ke saath chipakta); Worm = **bhagdu** (khud daudta hai).

4.3 Flowchart

FLOWCHART



4.4 Virus vs Worm (Table)

Basis	Virus	Worm
Host chahiye?	Haan (file/program)	Nahi (standalone)
Failta kaise	File run hone par	Khud network me
Speed	Dheema (user action)	Bahut tez (automatic)

Bandwidth	Kam effect	Network saturated
Remove	File se delete	Network clean karna

4.5 Types (Ek Nazar)

Type	Kya karta hai
Boot sector virus	Startup me load hota hai
Macro virus	Office documents ke macro se
File infector	EXE/application files ko infect
Polymorphic virus	Har baar apna code badalta hai (detection mushkil)
Resident virus	Memory me reh kar failta hai

4.6 Prevention (Bachao Ke Tarike)

- **Antivirus updated** rakho + regular scan.
- **Email attachments** bina check kholo mat.
- Software **official sources** se lo.
- **USB/devices** scan karke use karo.
- Network par **firewall + endpoint security**.
- **Backup** — virus delete karke bhi data safe rahe.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Virus = host file me attach + file run par failta.
2. Worm = self-replicating, network me khud failta.
3. Virus slow + files corrupt; worm bandwidth + systems khatam.
4. Bachao = antivirus + attachments se savdhaan + backup.

Viva Q1. Virus aur worm me main farak?

Ans: Virus ko host file chahiye, worm khud network me failta hai.

Viva Q2. Polymorphic virus kya hai?

Ans: Har baar apna code badalta hai — antivirus ko confuse karta hai.

MEMORY TRICK

Virus = Chipku, Worm = Bhagdu. Trick: Virus me **host** = ghar — bina ghar ke nahi rahta; Worm me **W = Web** — web par daudta hai.

PART 5 – TROJAN HORSES

Trojan Horse — malware jo **accha/legal software ban ke** aata hai — andar danger. Naam usi purani kahaani se hai — ladai wala ghoda.

5.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek app thi — **‘Free Games 2026’** — bilkul sahi dikhne wali. Rohit ne download ki.

Game chali bhi — par sath-sath:

- Uske **bank app** ka data nikal raha tha.
- **Remote access** ka gate khul gaya — hacker Rohit ka phone control kar sakta tha.
- Uske **contacts** me fake messages gaye.

Rohit ko lagta raha — ‘bas ek game hai’.

Yahi trojan ka khel hai — bahar se sweet, andar se nuksan.

5.2 Definition

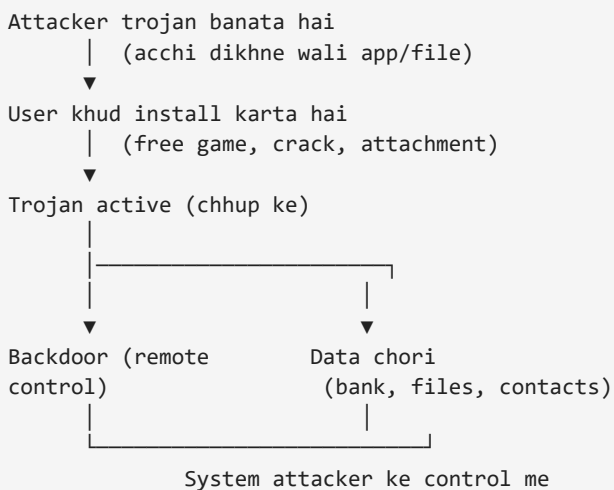
DEFINITION

Trojan Horse (Trojan) — malware jo **legitimate/useful program ke roop me chhupa** hota hai — user khud install karta hai — aur andar se **backdoor, data chori, remote access** karta hai.

Simple: Ghoda dakhil karo, **sipahi andar se nikle** — bilkul Trojan war wali story.

5.3 Flowchart

FLOWCHART



5.4 Types (Table)

Type	Kya karta hai
Remote Access Trojan (RAT)	Attacker ko system ka full control deta
Banking trojan	Bank pages fake bana kar credentials le
Keylogger trojan	Keystrokes record karta
Backdoor trojan	Hidden entry khulta
DDoS trojan	Botnet ka hissa banata

FakeAV trojan	Fake antivirus dikha kar paisa/access leta
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5.5 Trojan vs Virus vs Worm (Table)

Basis	Trojan	Virus	Worm
Failta kaise	Khud failta nahi — user install karta	Host file se	Network me khud
Replicate?	Nahi	Haan	Haan (fast)
Main khatra	Backdoor/data chori	Files damage	Bandwidth + spread

5.6 Prevention (Bachao Ke Tarike)

- **Free apps/cracks** se bacho — trojan ka favourite rasta.
- Apps **official store** se hi install karo.
- **Permissions** dekho — game ko camera/contacts kyu chahiye?
- **Antivirus** + app verification ON rakho.
- Attachment/link jo **unsolicited** ho — na kholo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Trojan = acchi dikhne wali app/file jisme malware chhupa hai.
2. Ye khud failta nahi — user khud install karta hai.
3. Types = RAT, banking, keylogger, backdoor, DDoS trojan.
4. Bachao = official store + permissions + antivirus.

Viva Q1. Trojan ka naam kyu hai?

Ans: Trojan war wale ghode se — bahar accha, andar danger.

Viva Q2. RAT kya hai?

Ans: Remote Access Trojan — attacker ko system ka remote control deta hai.

MEMORY TRICK

Trojan = Ghar me aaya mehmaan, andar se loot. Trick: **Trojan = Tarike se ghusna** — T = Trusted dikhna, phir loot.

PART 6 – BACKDOORS

Backdoor — system ka **chhupa hua gate** — jisse attacker jab chaahe andar aa jata hai, bina normal login ke.

6.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Hacker ne pehli baar ek website ki **weak password** se login kiya aur kuch data nikala. Phir usne site me ek **backdoor** install kar diya — ek chhupa hua URL/function.

Website wale ne password badal liya, firewall laga diya — par:

- Hacker ab bhi **backdoor se andar** aa jaata hai.
- Har baar kuch na kuch le jaata hai.
- Backdoor ko dhoondhna **bahut mushkil** hota hai.

Ye chhupa hua gate hi Backdoor hai — aur cyber security ki sabse badi gaddari.

6.2 Definition

DEFINITION

Backdoor — system me chhupa hua **secret entry point** — jo normal authentication ko **bypass** karke attacker ko access deta hai — ya to developer ne chhoda, ya attacker ne install kiya.

Simple: Ghar ka **chhupa hua darwaza** — jo sirf chor jaanta hai.

6.3 Flowchart

FLOWCHART

Backdoor kaise aata hai?



6.4 Types (Table)

Type	Kya hota hai
Software backdoor	Malware/trojan se install
Hardware backdoor	Device/chip me built-in
Default credentials	admin/admin, root chhoda hua
Hidden accounts	Chhupa hua user account
Web backdoor	Website me chhupa hua script (webshell)

6.5 Important Points

IMPORTANT POINTS

1. Backdoor **authentication bypass** karta hai — password ki zaroorat nahi.
2. Detecting backdoor **mushkil** — kyunki wo normal traffic jaisa lagta hai.
3. **Rootkits** backdoor ko aur bhi chhupa dete hain.
4. Backdoor milne par poori **forensic investigation** + system rebuild chahiye.
5. Good practice: **default passwords + hidden services** check karna.

6.6 Prevention (Bachao Ke Tarike)

- **Default passwords** turant badlo.
- **Patch management** — exploits se backdoor na ban paye.
- Network **monitor** karo — suspicious connections dekho.
- **File integrity** check (hash) — chhupa script pakdo.
- Antivirus + **EDR** (endpoint detection) rakho.
- Backdoor milne par **system clean/reinstall** karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Backdoor = chhupa hua secret entry point — authentication bypass.
2. Aata kaise = attacker installs ya vendor chhode (default creds).
3. Detection mushkil; rootkit aur chhupata hai.
4. Bachao = default password badlo + patching + monitoring + hash check.

Viva Q1. Backdoor se kya hota hai?

Ans: Attacker bina normal login ke baar-baar system me ghus sakta hai.

Viva Q2. Backdoor kaise install hota?

Ans: Trojan/exploit se attacker install karta ya vendor ki default/galt setting se.

MEMORY TRICK

Backdoor = Chhupa hua darwaza. Trick: **Backdoor = Back ka door** — aage ka darwaza (login) lock, par pichwale se andar.

PART 7 – DoS AND DDoS ATTACKS

DoS — ek source se itna traffic ki website/service band ho jaye. **DDoS** — hazaaron computers (botnet) se saath-me attack — rokna mushkil.

7.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Diwali sale ke din ek **online shopping site** par lagaathaar orders aa rahe the.

Ek hacker group ne:

- Hazaaron **hacked computers** (botnet) ko control me liya.
- Sab computers ne **ek saath** site par fake requests bheji.
- Server ka traffic **itna badh gaya** ki:
- Real customers ka site **khulna band** ho gaya.
- Server crash ho gaya — site band.

Site ko 2 din lag gaye wapas chalane me — **crore ka nuksan**.

Bahut saare computers ka ek-saath attack — DDoS. Ek hi computer ka — DoS.

7.2 Definition

DEFINITION

DoS (Denial of Service) — ek system/source se itne requests bhejna ki target server/service **overload ho kar legit users ko service na de** sake.

DDoS (Distributed DoS) — **bahut saare computers (botnet)** se distributed attack — ek source se nahi, isliye block karna aasaan nahi.

Simple: DoS = ek aadmi darwaza tod raha; DDoS = poori bheed darwaza tod rahi.

7.3 Flowchart

FLOWCHART

DoS: Attacker ----1 system----> Target
single source, traffic flood

DDoS: Botnet (hazaaron hacked PCs)



Target server (overload)



Server down / legit users out

7.4 DoS vs DDoS (Table)

Basis	DoS	DDoS
Source	Ek system/connection	Hazaaron systems (botnet)
Block karna	Ek IP block karo	Mushkil — thousands IPs
Speed	Dheema possible	Tez aur powerful
Detect	Aasaan (single source)	Mushkil
Example	SYN flood ek IP se	Botnet ka attack

7.5 Types / Techniques (Table)

Type	Kya hota hai
SYN flood	Half-open connections bana kar server ko bekar karna
Ping of Death	Bade/badle ping packets se system crash
Smurf attack	Spoofed ping broadcast se network flood
UDP flood	UDP packets se bandwidth khatam
HTTP flood	Site par fake requests (application layer)
Botnet DDoS	Hacked devices se distributed attack

7.6 Prevention (Bachao Ke Tarike)

- **Rate limiting** — ek IP se limited requests.
- **Firewall / IPS** — abnormal traffic block.
- **CAPTCHA** — fake bots rokna.
- **CDN / DDoS protection** service (Cloudflare jaisi).
- **Server load balancing** + extra capacity.
- **Monitoring + alert** — attack shuru hote hi action.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. DoS = ek source se overload — service band.
2. DDoS = botnet se distributed attack — block mushkil.
3. Types = SYN flood, ping of death, smurf, UDP flood, HTTP flood.
4. Bachao = rate limit, firewall/IPS, CDN protection, monitoring.

Viva Q1. DoS aur DDoS me farak?

Ans: DoS ek source se; DDoS hazaaron sources (botnet) se.

Viva Q2. SYN flood me kya hota?

Ans: Half-open connections bana kar server ki capacity khatam karna.

MEMORY TRICK

DoS = Ek aadmi, DDoS = Bheed. Trick: DDoS me **D = Distributed = Duniya bhar ke** computers. Jaise bheed-bhad me darwaza koi nahi rok paata.

PART 8 – BUFFER OVERFLOW

Buffer Overflow — ek **chhote dabbe (buffer)** me zyada data daal diya — aur data **bahar leak** hoke program ki memory ko corrupt kar deta hai — attacker apna code chala leta hai.

8.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek website ke **login page** me username box sirf **20 letters** rakhta tha.

Hacker ne box me **200 letters** daal diye — usme se kuch letters **malicious code** the.

- Data buffer se **overflow** hokar memory ke doosre hisse me gaya.
- Program ne **attacker ka code** execute kar diya.
- Hacker ko **remote control** mil gaya.

Ye sab us program ki galti se hua — usne input ki **lambaai check nahi ki**.

Zyada input se memory corrupt + attacker code — Buffer Overflow.

8.2 Definition

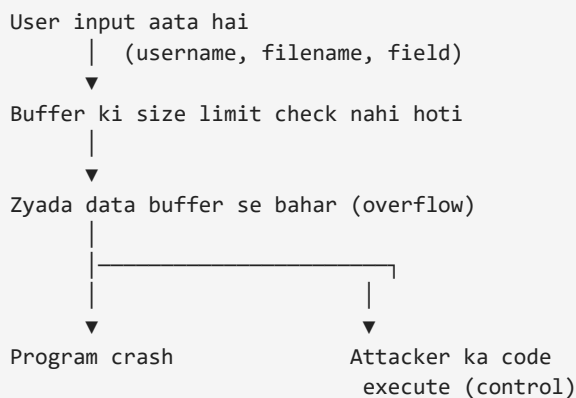
DEFINITION

Buffer Overflow — program ne ek fixed-size **buffer (memory area)** me capacity se **zyada data** likh diya — data adjacent memory me phail jata hai — program crash karta ya **attacker apna code execute** karwa leta hai.

Simple: Glass me paani se **zyada paani** daala — sab taraf phail gaya.

8.3 Flowchart

FLOWCHART



8.4 Types (Table)

Type	Kya hota hai
Stack overflow	Stack me buffer overflow — return address badalna
Heap overflow	Heap memory me overflow — data structures corrupt
Integer overflow	Bade number se variable bahar
Format string	Galat format se memory read/write

8.5 Important Points

IMPORTANT POINTS

1. Overflow tab hota hai jab program **input ki length validate nahi** karta.
2. Attacker overflow se **code injection** karta hai — system control.
3. **Stack overflow** me return address hijack hota hai — sabse common.
4. Ye **C/C++** me zyada hota hai (bina bound check); modern languages safe.
5. Patch lage na ho to **known buffer overflow** sabse khatarnak.

8.6 Prevention (Bachao Ke Tarike)

- **Input validation** — length aur type check karo.
- **Safe functions** (strcpy_s, gets nahi) use karo.
- **ASLR + DEP** — memory protection ON.
- **Compiler protections** (stack canary).
- Software **patch/update** karte raho.
- Code **review + testing** (fuzzing) karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Buffer overflow = zyada input, buffer se bahar, memory corrupt.
2. Attacker code execute karwa leta hai (code injection).
3. Common = stack overflow (return address hijack).
4. Bachao = input validation + ASLR/DEP + safe functions + patching.

Viva Q1. Buffer overflow me kya galat hota hai?

Ans: Program input ki length check nahi karta — data buffer se bahar jata hai.

Viva Q2. Stack overflow me kya hijack hota hai?

Ans: Return address — attacker apna code wahan daal deta hai.

MEMORY TRICK

Buffer Overflow = Paani ka glass zyada bharna. Trick: Buffer = **Bucket** — bucket chhoti, paani zyada — bahar phaila. Isliye hamesha **check input**.

PART 9 – ATTACK ON WIRELESS NETWORKS

WiFi par attack — kyunki WiFi ke signals **hawa me hote hain** — koi bhi paas khada ho kar pakad sakta hai. Weak security to bhi asaan ho jati hai.

9.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek cafe ki WiFi password thi — **WEP** (purani security). Hacker ne:

- **Packets sniff** kiye — WiFi ke signals me data pakda.
- WEP ki **weakness** use kar ke password tod diya.
- WiFi se jud kar **network me ghusa** — aur doosre devices ka data dekha.
- Ek **fake WiFi (evil twin)** bhi bana diya — 'Cafe_FreeWiFi' naam ka — jisse logon ke login churata raha.

Hawa me signal, asaan me attack — Wireless Network Attack.

9.2 Definition

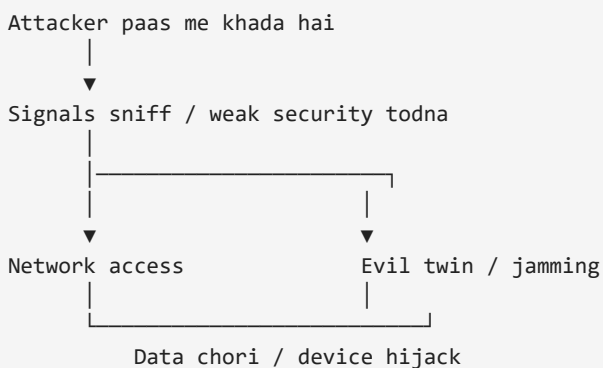
DEFINITION

Wireless Network Attack — WiFi ya wireless network ke **signals/security ko break** karna — password cracking, sniffing, fake access point, jamming — taaki data pakda jaye ya network me ghusa jaye.

Simple: Wired me taaar kaatna padta; wireless me **hawa hi rasta** hai.

9.3 Flowchart

FLOWCHART



9.4 Types of Wireless Attacks (Table)

Attack	Kya hota hai
Packet sniffing	Signals se data capture
WEP/WPS cracking	Weak security todna (WPS pin brute force)
Evil twin	Fake WiFi same naam — login churana
Rogue access point	Bina permission ka WiFi point
Deauthentication	Users ko WiFi se gira kar data pakadna
Jamming	Signals me interference — network band
KRACK	WPA2 ke handshake me vulnerability

9.5 Important Points

IMPORTANT POINTS

1. **WEP purani aur weak** hai — minutes me toot jati hai.
2. **WPA2/WPA3** strong hai — par WPS off rakho.
3. Public WiFi par **data plain** ja sakta hai — VPN use karo.
4. **Evil twin** same naam ka fake network — login churata hai.
5. Wireless attack ke liye attacker ko **network ke paas** hona padta hai.

9.6 Prevention (Bachao Ke Tarike)

- **WPA3/WPA2** use karo — WEP kabhi nahi.
- **WPS band** rakho.
- Router ka **default password** badlo.
- Public WiFi par **VPN** use karo.
- **SSID broadcast** + MAC filtering se thoda bachav.
- Router **firmware update** karte raho.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Wireless attack = signals/security break karna.
2. Types = sniffing, WEP cracking, evil twin, rogue AP, deauth, jamming.
3. WEP weak; WPA2/WPA3 strong; WPS off rakho.
4. Public WiFi par VPN + sensitive kaam na karo.

Viva Q1. Evil twin attack kya hai?

Ans: Same naam ka fake WiFi banakar login/data churana.

Viva Q2. WEP kyu kharab hai?

Ans: Purani aur weak security — kuch hi minutes me toot jati hai.

MEMORY TRICK

Wireless = Hawa ka network. Trick: **WiFi = Wide-open** — signal hawa me hai, isliye hamesha **WPA2/3 + VPN**. Evil twin = **nakli bhai**.

PART 10 – PHISHING (METHOD AND TECHNIQUES)

Phishing — asli bank/company ban ke **fake link** bhejna — victim login kare to uske **credentials** churana. Ye sabse zyada successful attack hai.

10.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek aadmi ko email aayi:

- Sender: **support@hdfc-bank.com** (dhyan se dekho — dash hai, asli nahi).
- Message: “Aapka account block ho jayega. **Turant verify karo**” + link.

Usne link kholi — page bilkul **bank jaisa** dikhta tha. Usne **user + password + OTP** daal diye.

- Hacker ke paas saare **credentials** chale gaye.
- Account se paise nikal liye.

Fake page/link se credentials churana — Phishing.

10.2 Definition

DEFINITION

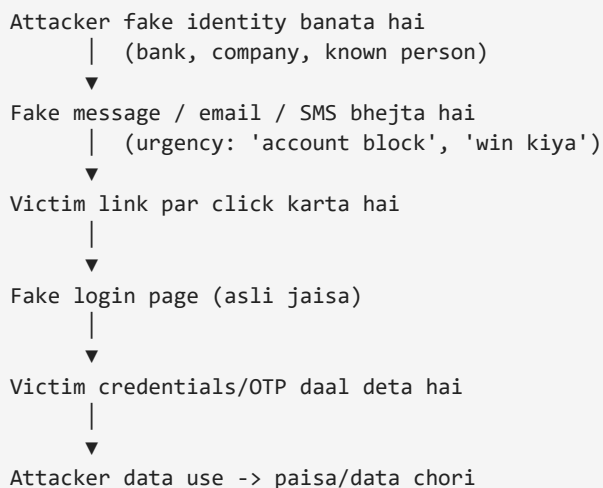
Phishing — ek **social engineering** attack jisme attacker **trusted entity (bank, company, person)** ban ke fake message/page bhejta hai — victim apne **login, OTP, card details** khud de deta hai.

Method: Fake message → Fake link → Fake page → Credentials → Nuksan.

Simple: Machhli pakadne ke liye **chara** (fake link) dalna — phishing = fishing.

10.3 Flowchart – Method

FLOWCHART



10.4 Phishing Techniques (Table)

Technique	Kya hota hai
Email phishing	Fake email + link
Spear phishing	Ek specific person ko targeted attack
Whaling	Bade official (CEO) ko targeted
Smishing	SMS ke through phishing

Vishing	Phone call ke through (voice phishing)
Pharming	DNS me chhed — asli site par bhi fake page bhi
Clone phishing	Asli email ki copy + badli hui link
Link manipulation	Galat URL — hdfc-bank.com jaisa nakli
Tab nabbing	Tab replace karke login churana

10.5 Important Points

IMPORTANT POINTS

1. Phishing **technique par nahi** — **bharosay par** chalti hai — isliye awareness sabse bada bachav.
2. **Urgency** wale messages par turant mat click karo.
3. **URL hamesha check** karo — asli vs nakli (dash, extra letter).
4. Pharming me DNS badal jata hai — URL sahi dikhne par bhi fake page.
5. Bank **kabhi OTP/ password email-call se nahi maangta**.

10.6 Prevention (Bachao Ke Tarike)

- Links par click karne se pehle **URL verify** karo.
- **2FA** — credentials churi hone par bhi guard.
- Bank/company ko **direct official channel** se call karo.
- **Email security** — sender address + spelling dekho.
- Unknown **attachments** na kholo.
- **Antivirus/anti-phishing** browser protection ON.
- Kisi ne credentials de diye to **turant password + bank card** change/block karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Phishing = fake trusted entity ban ke credentials churana.
2. Method = fake message + link + page + credentials.
3. Techniques = email, spear, whaling, smishing, vishing, pharming, clone.
4. Bachao = URL check + 2FA + urgency se savdhaan.

Viva Q1. Spear phishing kya hai?

Ans: Ek specific person (targeted) ke liye personal phishing attack.

Viva Q2. Pharming me kya badalta hai?

Ans: DNS — URL sahi dikhta hai par page fake hota hai.

MEMORY TRICK

Phishing = Fishing — fake link = **chara**, victim = machhli. Trick: **Urgency = Dhamki** — jo message 'turant karo' bole, socho phir click karo.

APPENDIX – QUICK REVISION SHEET (UNIT 5)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
Proxy Server	Beech ka server — asli IP chhupata hai
Anonymizer	Identity/IP fully chhupane wala tool
Password Cracking	Password recover/guess karna (dictionary, brute force)
Keylogger	Har keystroke record karta hai
Spyware	Activities par spy (browsing, screen, camera)
Virus	Host file me attach + file run par failta
Worm	Self-replicating — network me khud failta
Trojan Horse	Acchi app/file ban ke andar malware
Backdoor	Chhupa hua secret entry point (auth bypass)
DoS	Ek source se overload — service band
DDoS	Botnet (hazaaron systems) se distributed attack
Buffer Overflow	Zyada input buffer se bahar — memory corrupt + code run
Wireless Attack	WiFi signals/security break (evil twin, sniffing)
Phishing	Fake trusted entity ban ke credentials churana

Memory Tricks

- Proxy = **Paisa ka bichauliya** (IP nahi batata).
- Password = **PassWORD** — 123456 = khula darwaza.
- Keylogger = **KEY** in keyboard.
- Virus = **Chipku**, Worm = **Bhagdu** (W = Web).
- Trojan = **Ghar me mehmaan, andar loot**.
- Backdoor = **Back ka door** — pichhla darwaza.
- DDoS = **Distributed = Duniya bhar ke** computers.
- Buffer Overflow = **Bucket chhoti, paani zyada**.
- Wireless = **Hawa ka network** — WPA2/3 + VPN.
- Phishing = **Fishing** — fake link = chara.

Rapid Fire Viva (15 Questions)

Q	Answer
Proxy server se kya chhupata hai?	Asli IP
Brute force kya hai?	Har possible password combo try karna
Rainbow table kya hai?	Pehle se bani hash-value table
Keylogger kya record karta?	Keystrokes (passwords, OTP)

Virus ko kya chahiye?	Host file (attach hone ke liye)
Worm kaise failta?	Network me khud (bina host ke)
Trojan kaise install hota?	User khud install karta hai (accha ban ke)
Backdoor kya bypass karta?	Normal authentication
DoS vs DDoS?	DoS ek source, DDoS botnet se
SYN flood me kya hota?	Half-open connections se server bekar
Buffer overflow me kya check nahi hota?	Input ki length
WEP kyu weak hai?	Purani security, minutes me toot jati
Evil twin kya hai?	Same naam ka fake WiFi — login churana
Pharming me kya badalta hai?	DNS
Phishing ka target?	Credentials (login/OTP/card)

Note: Ye Unit 5 ka complete notes hai — cyber crime ke tools aur methods.